Supplementary material

Training parts versus matching parts in pose-based gait recognition

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Section, table and figure numbers without the prefix "S" refer to the main article. All values are Rank-1 in percent or Rank-1 differences in percentage points unless stated otherwise. Every table except Tables S1–S3 is generated by `make_tables.py` from `evidence_pr.json`, which `build_pr_evidence.py` assembles from the saved run artifacts.

S1 Three-seed factorial

The factorial trained six objectives with seeds 11, 22 and 33 under source revision `ad96b902...`; the five-seed cohort in the main text used revision `11b575d0...`. Both used the same recipe, identities and evaluation masks. Tables S1 and S2 report the factorial with the global and the learned projected-parts descriptors; its intervals use a crossed bootstrap with 100,000 draws. In this cohort part SupCon reduced the corrupted-probe aggregate by 6.12 points, $[-10.29, -2.00]$. The five-seed cohort estimates the same contrast at -1.27, $[-5.24, 2.74]$ (main Table 3), so the main text does not retain a harmful effect of part SupCon.

Table S3 lists training diagnostics. The part-triplet objective closely preserved clean global accuracy but its final normalized part variance was about 1.7×10^{-6}; at the selected epochs (60, 60 and 40) the variance averaged 2.03×10^{-6} across seeds and parts, and a nonzero finite part–triplet gradient cosine was available in only 9 of 96 scheduled measurements (the remainder had a zero-norm gradient for at least one compared loss). These observations indicate near-collapse of the local embeddings under this objective and recipe. Part SupCon avoided collapse (variance 3.34×10^{-3}), but in the combined model its gradient agreed less with the global triplet objective than the global SupCon gradient did (mean cosines 0.203 and 0.446). The cosines are positive, so the data do not show direct gradient opposition. Cosines are measured at the shared paired-view feature tensor, and variance diagnostics describe training batches rather than test embeddings.

Table S1: Three-seed factorial (earlier source revision). CASIA-B validation Rank-1 (%). Clean is mean ± seed standard deviation. "All > 0" averages four corruption families at severities 0.1, 0.3, 0.5, and 0.7. Occluded gallery averages the same positive-severity grid. Single-view averages NM, BG, and CL. Every cell uses three training seeds.

Global descriptor

Training objective	Clean	Clean gallery all > 0	Clean gallery $\rho = 0.5$	Occluded gallery all > 0	Single-view
Metric baseline	81.52 ± 2.43	63.90	64.25	67.93	67.36
Global SupCon	79.63 ± 3.23	63.23	64.15	67.42	66.18
Part SupCon	78.79 ± 1.14	57.78	55.57	64.26	65.33
Part triplet	81.27 ± 3.41	60.65	59.10	66.71	67.82
Global + part SupCon	80.05 ± 1.34	60.70	59.06	65.56	65.75
Global + part SupCon (ungated)	80.89 ± 1.59	61.72	60.59	66.99	66.29

Projected-parts descriptor

Training objective	Clean	Clean gallery all > 0	Clean gallery $\rho = 0.5$	Occluded gallery all > 0	Single-view
Metric baseline	83.04 ± 1.31	66.31	67.97	70.57	69.34
Global SupCon	82.79 ± 3.36	66.21	67.85	70.38	68.66
Part SupCon	75.42 ± 2.25	54.66	52.65	61.62	61.81
Part triplet	81.31 ± 3.42	60.66	59.11	66.73	67.82
Global + part SupCon	77.86 ± 4.45	57.48	54.84	63.18	63.40
Global + part SupCon (ungated)	78.20 ± 0.07	59.35	58.11	64.85	65.60

Table S2: Three-seed factorial. Paired Rank-1 difference from the metric baseline in percentage points (crossed 95% bootstrap interval over seed and validation identity). Negative values favor the baseline.

Training objective	Descriptor	Clean	Clean gallery, all > 0
Global SupCon	Global	-1.89 [-5.30, +1.05]	-0.67 [-4.86, +3.79]
Part SupCon	Global	-2.74 [-6.69, +1.18]	-6.12 [-10.29, -2.00]
Part triplet	Global	-0.25 [-5.22, +4.17]	-3.26 [-7.59, +1.06]
Global + part SupCon	Global	-1.47 [-5.85, +3.03]	-3.21 [-8.97, +2.89]
Global + part (ungated)	Global	-0.63 [-5.35, +5.30]	-2.19 [-5.35, +1.11]
Global SupCon	Projected parts	-0.25 [-3.79, +2.99]	-0.10 [-3.27, +3.08]
Part SupCon	Projected parts	-7.62 [-11.91, -3.03]	-11.65 [-16.02, -7.32]
Part triplet	Projected parts	-1.73 [-6.82, +2.90]	-5.65 [-9.85, -1.16]
Global + part SupCon	Projected parts	-5.18 [-10.61, +0.13]	-8.83 [-14.35, -2.29]
Global + part (ungated)	Projected parts	-4.84 [-8.75, -1.05]	-6.96 [-11.01, -3.08]

Training objective	Descriptor	Clean gallery, $\rho = 0.5$	Occluded gallery, all > 0
Global SupCon	Global	-0.10 [-6.13, +6.76]	-0.51 [-2.90, +1.76]
Part SupCon	Global	-8.68 [-15.55, -2.38]	-3.67 [-6.00, -1.27]
Part triplet	Global	-5.15 [-11.42, +1.28]	-1.22 [-4.29, +1.50]
Global + part SupCon	Global	-5.19 [-13.69, +4.17]	-2.37 [-6.16, +1.24]
Global + part (ungated)	Global	-3.65 [-8.91, +0.68]	-0.94 [-3.00, +1.05]
Global SupCon	Projected parts	-0.12 [-4.37, +4.10]	-0.20 [-2.13, +1.76]
Part SupCon	Projected parts	-15.32 [-22.19, -8.96]	-8.96 [-11.51, -6.03]
Part triplet	Projected parts	-8.86 [-15.80, -1.57]	-3.85 [-6.48, -1.32]
Global + part SupCon	Projected parts	-13.13 [-20.45, -4.51]	-7.39 [-11.19, -2.97]
Global + part (ungated)	Projected parts	-9.85 [-16.07, -4.53]	-5.73 [-8.08, -3.59]

Table S3: Three-seed factorial. Training diagnostics averaged over three seeds. Part variance is the mean coordinate-wise variance over the final 20 epochs. Cosines are measured every five epochs; parentheses give nonzero finite cosine measurements out of 96 scheduled.

Objective	Variance ($\times 10^{-3}$)	Part–global	Part–triplet
Metric baseline	1.269	–	–
Global SupCon	1.300	–	–
Part SupCon	3.338	–	0.212 (95)
Part triplet	0.002	–	0.012 (9)
Global + part SupCon	3.349	0.265 (96)	0.203 (96)
Global + part SupCon (ungated)	3.242	0.241 (96)	0.195 (96)

S2 Corruption calibration and evaluation matrix

Nominal severity ρ is the requested missing fraction. Whole-part masks remove complete anatomical regions and overshoot small requests, while temporal smoothing lowers the marginal dynamic-joint rate at small ρ and raises it at large ρ. Table S4 gives the removed fraction of real joint-frames, regenerated exactly as the evaluator drew each mask from its recorded seed, family order, realization and probe index. Figure S1 records where each family is used, and Figure S2 illustrates the four families.

Table S4: Mean removed fraction of observed joint-frames (%) at each nominal severity, CASIA-B development probes / Gait3D development probes.

Family	$\rho = 0.1$	$\rho = 0.3$	$\rho = 0.5$	$\rho = 0.7$
Random joint-frame	10.0 / 10.0	30.0 / 30.0	50.0 / 50.0	70.0 / 70.0
Dynamic joint	3.2 / 3.2	22.3 / 22.4	50.6 / 50.7	78.7 / 78.8
Whole part	18.8 / 18.9	35.3 / 35.3	49.4 / 49.3	70.6 / 70.6
Temporal interruption	10.0 / 10.0	30.0 / 30.0	50.0 / 50.0	70.0 / 70.0

Corruption family	**Training**	**Clean gallery**	**Occluded gallery**	**Positive ρ**
Random joint-frame	sampled	probe only	gallery + probe, independent	5 masks
Dynamic joint	held out	probe only	gallery + probe, independent	5 masks
Whole part	sampled	probe only	gallery + probe, independent	5 masks
Temporal interruption	sampled	probe only	gallery + probe, independent	5 masks

Severities: $\rho \in \{0, 0.1, 0.3, 0.5, 0.7\}$

Descriptors: global and part-aware; primary outcome: Rank-1

Figure S1: Evaluation matrix. Dynamic-joint corruption is held out from training. Positive-severity results average five deterministic corruption realizations, and gallery and probe corruptions are independently seeded in the occluded-gallery protocol.

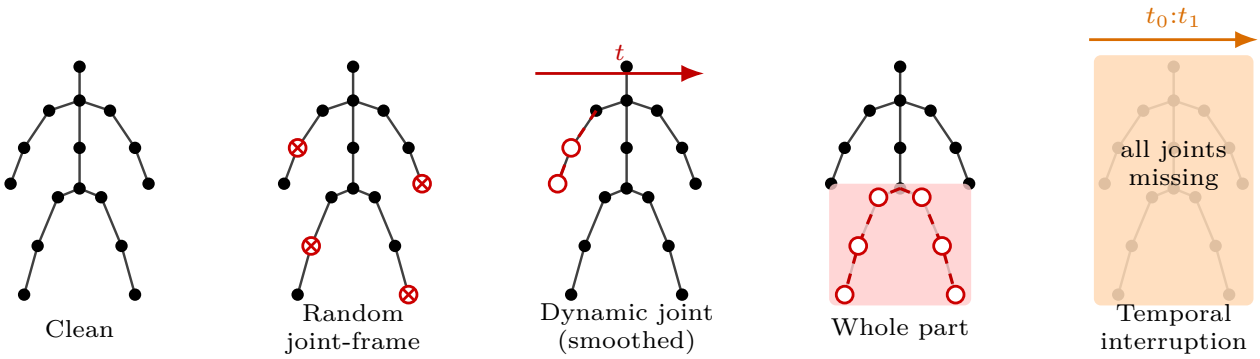


Figure S2: Corruption families. Red marks unavailable joints or regions; orange marks a contiguous interval without pose. Random joint-frame masks vary independently, whereas dynamic-joint masks are temporally smoothed.

S3 Fusion-weight sensitivity

The local-score weight λ was declared as 0.5 before any comparison. Table S5 shows the reliability-weighted raw-part matcher on the frozen metric-baseline checkpoints at all five evaluated weights; $\lambda = 0$ is the global descriptor. The best CASIA-B weight (0.75) would have removed the Gait3D gain, and part-only matching ($\lambda = 1$) is markedly worse than the global descriptor on Gait3D.

Table S5: Raw reliability-weighted part matching on frozen metric-baseline checkpoints at different local-score weights, Rank-1 (%).

	CASIA-B, five seeds		Gait3D, three seeds	
Local-score weight λ	Clean	Corr. probe	Clean	Corr. probe
0.0	79.60	60.93	35.89	23.96
0.25	81.19	62.97	36.67	25.26
0.5 (declared)	82.40	64.66	36.89	25.81
0.75	83.43	65.71	33.67	23.87
1.0	84.37	65.20	21.33	15.07

S4 Raw matching for every trained objective

Table S6 lists the values plotted in main Figure 3: raw reliability-weighted part matching minus global matching from the same checkpoints, with crossed seed-by-identity bootstrap intervals (20,000 draws).

Table S6: Raw reliability-weighted part matching minus global matching (Rank-1 points, crossed 95% intervals), from each training run's own evaluation. The checkpoint-control analysis in main Table 4 re-encodes the metric-baseline checkpoints and differs from these values by at most 0.05 points.

Cohort	Training objective	Clean	Corr. probe	Corr. gallery
CASIA-B, 5 seeds	CE, no masking	+2.53 [0.61, 4.75]	+2.32 [1.47, 3.24]	+3.50 [2.60, 4.34]
CASIA-B, 5 seeds	Masked CE	+2.85 [1.26, 4.39]	+3.48 [2.23, 4.63]	+3.54 [2.72, 4.33]
CASIA-B, 5 seeds	Matched CE	+1.44 [−0.18, 3.23]	+3.21 [1.74, 4.69]	+2.61 [1.71, 3.60]
CASIA-B, 5 seeds	Metric baseline	+2.75 [0.38, 4.75]	+3.73 [1.67, 5.68]	+3.66 [2.12, 5.06]
CASIA-B, 5 seeds	Global SupCon	+2.68 [0.88, 4.70]	+4.28 [2.79, 5.92]	+3.83 [2.87, 4.81]
CASIA-B, 5 seeds	Part SupCon	+1.89 [0.18, 3.69]	+2.87 [1.32, 4.31]	+2.70 [1.96, 3.41]
CASIA-B, 5 seeds	Global + part SupCon	+1.94 [0.48, 3.54]	+2.95 [1.96, 4.10]	+2.73 [2.06, 3.49]
Gait3D dev., 3 seeds	CE, no masking	−1.00 [−3.78, 1.67]	+0.90 [0.49, 1.33]	+0.86 [0.46, 1.26]
Gait3D dev., 3 seeds	Matched CE	+0.33 [−2.33, 2.89]	+2.25 [1.10, 3.43]	+1.64 [0.70, 2.62]
Gait3D dev., 3 seeds	Metric baseline	+1.00 [−2.56, 4.67]	+1.85 [0.76, 2.98]	+1.57 [0.71, 2.50]
Gait3D dev., 3 seeds	Global + part SupCon	+1.44 [−1.00, 4.00]	+2.02 [1.03, 3.01]	+1.72 [0.95, 2.50]
Gait3D test, 5 seeds	CE, no masking	+1.46 [0.44, 2.47]	+0.60 [0.44, 0.76]	+0.71 [0.52, 0.90]
Gait3D test, 5 seeds	Metric baseline	+1.99 [0.61, 3.38]	+1.59 [1.07, 2.12]	+1.19 [0.76, 1.63]

S5 Additional development results

Table S7 breaks the five-seed CASIA-B single-view average of main Table 2 into its normal, bag and clothing conditions. Table S8 gives the Gait3D development results by objective (300 held-out training identities, one probe each, three seeds).

Table S7: CASIA-B development, single-view-gallery protocol excluding identical views, five seeds (mean ± seed standard deviation).

	Global descriptor			Raw parts, reliability-weighted		
Training objective	NM	BG	CL	NM	BG	CL
CE, no masking	76.77 ± 1.40	58.30 ± 2.40	53.29 ± 3.69	80.80 ± 1.67	61.69 ± 3.43	56.96 ± 4.06
Masked CE	65.27 ± 3.37	46.11 ± 2.51	39.98 ± 3.75	69.30 ± 2.84	48.61 ± 2.34	41.90 ± 3.29
Matched CE	79.99 ± 3.59	62.01 ± 3.23	54.48 ± 1.40	83.47 ± 2.96	63.81 ± 2.71	56.77 ± 2.73
Metric baseline	79.52 ± 1.38	59.25 ± 3.21	57.56 ± 2.80	83.20 ± 1.20	62.05 ± 1.98	59.83 ± 1.83
Global SupCon	79.42 ± 2.06	61.23 ± 1.20	59.25 ± 3.07	83.52 ± 1.37	63.97 ± 1.33	61.17 ± 3.53
Part SupCon	77.97 ± 1.07	61.11 ± 2.83	53.02 ± 2.25	81.33 ± 1.23	62.89 ± 1.82	54.52 ± 3.00
Global + part SupCon	78.18 ± 2.50	59.68 ± 2.74	58.48 ± 1.54	81.11 ± 1.72	62.34 ± 3.18	60.49 ± 1.48

Table S8: Gait3D development Rank-1 (%), three seeds, 300 probes (mean ± seed standard deviation where shown).

	Global descriptor			Raw parts, reliability-weighted	
Training objective	Clean	Corr. probe	Corr. gallery	Clean	Corr. probe
CE, no masking	28.44 ± 2.55	8.26 ± 0.50	9.58	27.44	9.15
Matched CE	31.56 ± 1.92	21.50 ± 0.92	19.24	31.89	23.74
Metric baseline	35.89 ± 2.55	23.96 ± 1.26	21.39	36.89	25.81
Global + part SupCon	34.44 ± 1.71	24.03 ± 0.64	21.84	35.89	26.05

S6 GPGait transfer: variants and fusion weights

Features were extracted with `scripts/fpg_part_matching.py`, which runs each trained GPGait checkpoint through FastPoseGait's own test loader, transforms and network code and records the 19 inference embeddings and the per-joint features of each branch's global backbone after temporal pooling. GPGait creates every graph layer's learnable adjacency from one shared tensor; the script copies each parameter before loading the checkpoint, as FastPoseGait's transfer to the GPU does, and refuses to run if parameters still share storage. With all 19 embeddings it reproduces the logged CASIA-B results (93.78, 79.89 and 69.12 against 93.77, 79.86 and 69.11) and Gait3D Rank-1 (23.13). Table S9 adds the exploratory variant that passes mean-plus-max region pools through each branch's trained whole-body projection and BNNeck layer (maximum reconstruction error of GPGait's own whole-body embeddings 7×10^{-4}) and fusion weights 0.25 and 0.75. No variant or weight lets raw parts improve on the full descriptor; at $\lambda = 0.25$ they change the whole-body descriptor by less than 0.5 points.

Table S9: GPGait with raw parts: declared and exploratory variants and other fusion weights (Rank-1 and mAP, %). Without raw parts, the full descriptor reaches 80.93 (CASIA-B average) and 23.13 / 16.79 (Gait3D Rank-1 / mAP), and the whole-body descriptor 71.85 and 17.95 / 12.59.

Descriptor	CASIA-B average	Gait3D Rank-1	Gait3D mAP
Declared: mean pool, no projection			
Raw parts only	16.20	9.40	6.20
Whole-body + raw	70.92	17.11	11.86
Full + raw	78.53	19.96	14.24
Whole-body + raw, $\lambda = 0.25$	72.00	17.53	12.55
Full + raw, $\lambda = 0.25$	80.70	22.28	15.64
Whole-body + raw, $\lambda = 0.75$	63.16	16.16	10.62
Full + raw, $\lambda = 0.75$	65.75	16.47	11.43
Exploratory: mean+max pool through the whole-body head			
Raw parts only	36.07	12.46	8.92
Whole-body + raw	71.29	16.58	11.83
Full + raw	77.61	18.90	13.37
Whole-body + raw, $\lambda = 0.25$	72.21	17.63	12.38
Full + raw, $\lambda = 0.25$	80.59	21.65	15.34
Whole-body + raw, $\lambda = 0.75$	62.91	14.57	10.53
Full + raw, $\lambda = 0.75$	64.55	15.84	11.07

S7 Detector-confidence stratification on Gait3D

Gait3D has no occlusion annotation. Each development probe is described by proxies computed only from its released HRNet confidences over the real frames the evaluator embeds: the fraction of joint-frames with confidence below 0.5 (primary), the fraction below 0.3, one minus the mean confidence, and one minus the lowest mean part confidence. The primary proxy, threshold and endpoints were fixed from the confidence and height distributions before outcomes were examined; at threshold 0.3 a fifth of probes have no qualifying joint-frame. The primary endpoint is the height-matched interaction: within each skeleton-height tertile, probes are split into tertiles of the primary proxy, the raw-minus-global Rank-1 gain among the most low-confidence probes minus the gain among the least is computed, and the three height strata are averaged with equal weight. The secondary endpoint repeats the contrast over global tertiles. Intervals resample training seeds and probe identities as crossed clusters (20,000 draws). The tertile edges of the primary proxy are 1.65%

and 4.28% of joint-frames (range 0.09–19.7%), and its Spearman correlations with sequence length and skeleton height are -0.05 and -0.29. The primary endpoint is -0.68 points, $[-7.69, 6.22]$, and the secondary $+1.00$ points, $[-6.54, 8.45]$. Low confidence also follows resolution and blur, bins use probe evidence only, and gallery tracks may themselves be occluded; the analysis therefore cannot establish whether the matching gain addresses real occlusion.

S8 Test-set details

Main Table 5 reports means. Tables S10 and S11 add seed standard deviations, the uniform-weight raw descriptor, Rank-5, Rank-10, mINP and parameter counts, and Table S12 gives the paired intervals for the GPGait descriptor variants.

Table S10: CASIA-B official test (identities 075–124), standard single-view protocol excluding identical views, Rank-1 (%). Our rows give mean ± standard deviation over five seeds of the metric baseline; reproduced baselines were trained once with official FastPoseGait configurations, and reported values are those of the FastPoseGait model zoo. Parameters include each model's identity classifier.

System	NM	BG	CL	Average	Params. (M)
Ours, global	67.64 ± 2.83	49.50 ± 1.92	27.91 ± 0.94	48.35 ± 1.85	1.69
Ours, raw (uniform)	74.02 ± 3.04	55.56 ± 2.04	32.54 ± 1.88	54.04 ± 2.21	1.69
Ours, raw (reliability)	74.14 ± 3.04	55.47 ± 2.01	32.52 ± 1.81	54.04 ± 2.18	1.69
GaitGraph2 (reproduced)	76.66	67.23	60.63	68.17	0.35
GaitGraph2 (reported)	80.29	71.40	63.80	71.83	–
GPGait (reproduced)	93.77	79.86	69.11	80.91	3.65
GPGait (reported)	93.60	80.15	69.29	81.01	–

Table S11: Gait3D test, available-pose subset (947 probes, 5,122 gallery sequences), evaluated once. Our rows give means over five seeds (± seed standard deviation for Rank-1 and mAP). "Metric" is the metric baseline and "raw" the raw-part descriptor. Parameters include the 2,892-way identity classifier where present.

System	Rank-1	Rank-5	Rank-10	mAP	mINP	Params. (M)
CE, global	15.48 ± 1.38	30.41	38.46	12.07 ± 0.40	7.16	2.41
CE, raw (reliability)	16.94 ± 1.26	31.89	39.64	12.87 ± 0.51	7.91	2.41
Metric, global	21.88 ± 1.18	38.94	46.99	16.89 ± 0.43	10.17	2.41
Metric, raw (uniform)	23.86 ± 0.64	39.94	47.79	17.82 ± 0.24	10.90	2.41
Metric, raw (reliability)	23.86 ± 0.68	40.02	47.69	17.79 ± 0.24	10.86	2.41
GaitGraph2 (reproduced)	8.87	24.50	31.47	8.38	4.77	0.35
GPGait (reproduced)	23.13	36.43	44.14	16.78	10.51	18.11

Table S12: Raw part matching applied to GPGait, with paired differences. "Full" is the published descriptor and W its three whole-body embeddings; raw parts are fused at $\lambda = 0.5$ on GPGait's native distance. Intervals resample test identities.

	CASIA-B official test				Gait3D test	
GPGait descriptor	NM	BG	CL	Average	Rank-1	mAP
Full (19 embeddings)	93.78	79.89	69.12	80.93	23.13	16.79
Whole-body only (W)	87.86	68.21	59.49	71.85	17.95	12.59
Raw parts only	22.12	14.85	11.62	16.20	9.40	6.20
W + raw parts	86.53	67.15	59.07	70.92	17.11	11.86
Full + raw parts	91.55	77.01	67.05	78.53	19.96	14.24
Paired differences (points), 95% interval over test identities						
W − full	−9.08 [−10.71, −7.43]				−5.17 [−7.29, −3.06]	−4.19 [−5.31, −3.09]
Full + raw − full	−2.40 [−3.29, −1.53]				−3.17 [−4.65, −1.80]	−2.54 [−3.23, −1.88]
W + raw − W	−0.94 [−1.77, −0.15]				−0.84 [−2.11, 0.42]	−0.73 [−1.32, −0.18]

S9 Reproduced baselines and provenance

FastPoseGait baselines GPGait and GaitGraph2 were run with the official FastPoseGait configurations (`gpgait`, `gaitgraph2` and their Gait3D variants), changing only data paths, run names and intermediate testing (disabled; the final checkpoint is evaluated). Poses were exported from the same data files as our runs without our normalization. For Gait3D, frames are stacked in numeric frame-ID order (the official pretreatment sorts file names lexicographically, which differs only where frame IDs change digit count), HRNet confidences above one were clipped to one, and only the available-pose subset is present. The FastPoseGait source snapshot has fingerprint `4fddc43c...`. Parameter counts are those printed by the framework.

Hashes The five-seed CASIA-B cohort, both Gait3D stages and both descriptor-control cohorts record source SHA-256 `11b575d0...`. Dataset checksums, which include protocol metadata, are `123e838c...` (CASIA-B development), `001c9427...` (Gait3D development) and `132636db...` (Gait3D final). The CASIA-B official-test runs record source `7a351cac...`, dataset `9823d021...` and configuration `f25bd174...`; the three-seed factorial and its controls record sources `ad96b902...` and `eddf23a2...`. Runs record source hashes rather than repository commits.

Hardware All training and timing used NVIDIA H100 NVL multi-instance GPU slices (1g.12gb, 2g.24gb and 3g.47gb profiles). Latency in main Section 5.6 was measured on a 1g.12gb slice after 100 warm-up passes, over 1,000 timed passes.

Regeneration From the project root, `pythonPaper/PatternRecognition/build_pr_evidence.py` reads the result directories listed in its header without modifying them, checks seeds, probe counts and hashes, and rewrites `evidence_pr.json`, main Figures 1, 3 and 4 and every generated table. It requires NumPy, PyYAML and Matplotlib and performs no training.